

HEF sensorHUB Documentation

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Section 1 HEF sensorHUB Application

HEF SensorHUB is a web platform built using React.js for the front end and powered by Flask (Python) for the back end. For every TUM account registered, this web application sets up two important components. First, it creates an instance of The Fraunhofer Open Source SensorThings-Server (FROST), which is a well-known open-source solution and the official reference implementation of the OGC SensorThings standard. This allows users to efficiently manage and work with sensor data. Additionally, HEF SensorHUB also provides an instance of Node-RED Essentials, an application designed for connecting hardware devices. It comes with a user-friendly, browser-based editor that simplifies the process of creating complex flows using various nodes available in the palette.

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Section 2 NodeRED, Grafana and FROST-Server

2.1 NodeRED

Node-RED is a powerful, open-source, and flow-based visual programming tool primarily used for simplifying the development of Internet of Things (IoT) applications. IoT involves connecting various devices and sensors to the internet to collect and exchange data, enabling automation, monitoring, and control in diverse contexts. Node-RED streamlines IoT development by offering a user-friendly interface where users can drag and drop nodes representing hardware interfaces, data processing functions, and communication protocols. This makes it exceptionally useful for IoT applications, where it can orchestrate the collection and processing of data from sensors, the execution of commands to actuators, and the integration with cloud services or other systems. It simplifies the creation of IoT solutions, reducing development time and the need for extensive coding, and allows developers to focus on building the logic and applications that drive real-world IoT scenarios.

2.2 Grafana

Grafana is an open-source platform used for monitoring systems and analyzing time-series data. Within the HEF sensorHUB ecosystem, it acts as the primary tool for transforming raw sensor observations into clear, real-time dashboards. Users can build interactive displays with various visual panels, such as graphs, heatmaps, and gauges. By using specialized plugins, such as the LinkSmart SensorThings Data Source, Grafana connects directly to the FROST-Server to retrieve and visualize metadata and sensor data via the OGC SensorThings API. Additionally, Grafana offers built-in alerting features to notify users if data exceeds specific thresholds, and it allows for the correlation of data from multiple sources to provide a more comprehensive view of your infrastructure.

2.3 FROST-Server

The Fraunhofer Opensource SensorThings-Server (FROST) is the first complete, open-source official reference implementation of the OGC SensorThings. The OGC SensorThings API provides an open, geospatial-enabled and unified way to interconnect the Internet of Things (IoT) devices, data, and applications over the Web. It provides a standard way to manage and retrieve observations and metadata from heterogeneous IoT sensor systems.

SensorThings API provides up-to-date measurement information such as:

- RESTful
- (Geo)JSON encoded
- Available via OASIS Odata URL patterns and query options
- Capable of directly including sensor data via the ISO MQTT protocol

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REST Cheat Sheet for SensorThings API:

	GET	POST	PATCH	DELETE
v1.1	Get index			
v1.1/Type	Get all of type	Create		
v1.1/Type(id)	Get one of type		Update	Delete
v1.1/Type(id)/Entity	Get linked entity			
v1.1/Type(id)/EntitySet	Get all linked	Create Linked		

Figure 1 - REST Cheat Sheet SensorThings API

Attributes:

1. Thing
2. Location
3. ObservedProperty
4. Sensor
5. Datastream
6. Observation

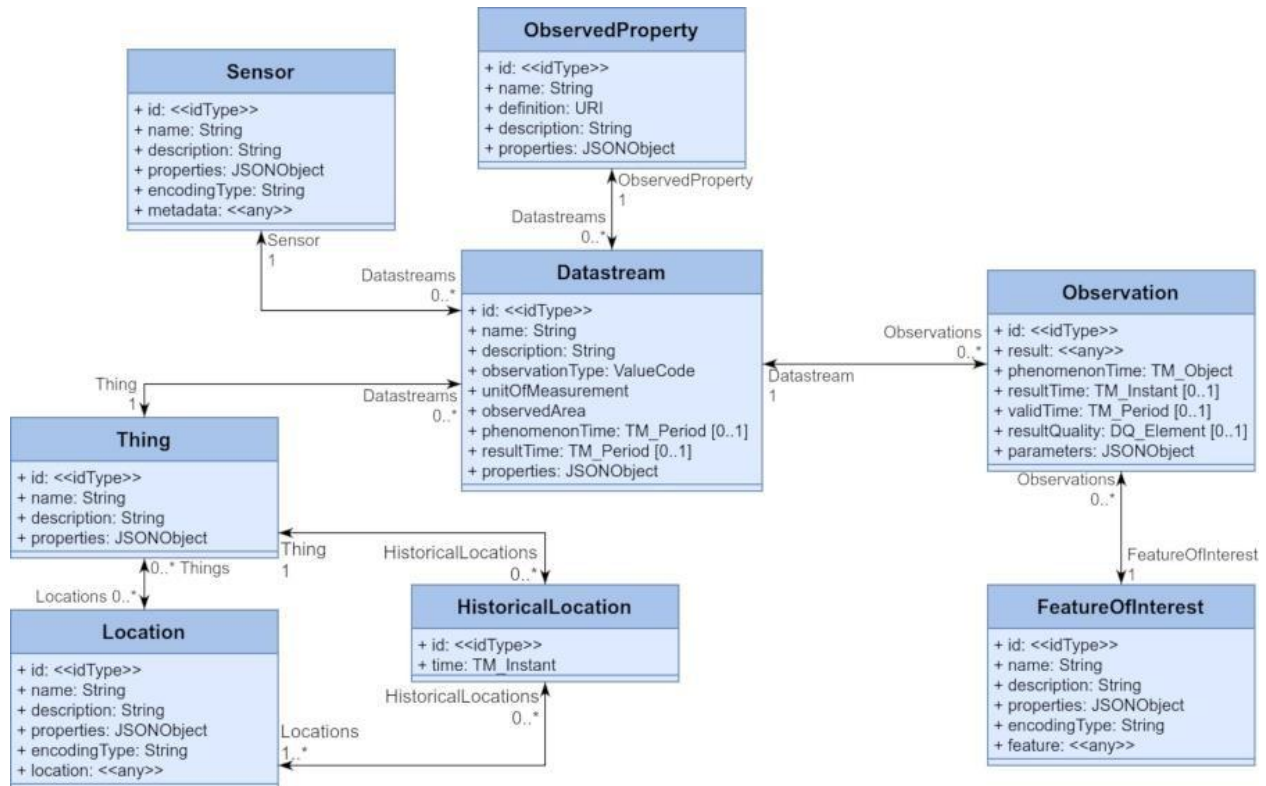


Figure 2 - SensorThings API Data Model

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Section 3 Application Component

3.1 Project Management

The project management dashboard acts as a central hub where users can oversee their collaborations, define specific user roles, and manage access requests. Within this interface, the "Own and Joined Projects" table provides a comprehensive overview of all active projects, displaying critical details such as the Group ID, project name, project description, and the owner of the project. From this table, you can streamline team management by approving or rejecting pending join requests and modifying individual member roles, such as the Reader role, to ensure proper access control.

For project creation and onboarding, users can utilize the "Create New Project" section by providing a unique project name and description. This section includes an optional "Include Node-RED" toggle, allowing you to incorporate flow-based IoT development tools if required for your specific project. If you wish to join an existing initiative, the "Join Existing Project" search bar allows you to locate projects by their Group ID and submit an access request. Once your request is sent, the project owner must review and approve it, which will then grant you full access to the project environment.

3.2 Devices

To access devices (referred to as "Things" in SensorThings terminology), you can navigate to the /devices page. Here, you will find a table displaying essential information such as the ID, name, and description of each device. You have the option to perform various actions on these devices, including updating or deleting them. Moreover, the real-time coordinates of these devices can be visualized on a map component.

In addition to the table view, you can visualize the real-time coordinates of these devices on a map component. This map features cluster functionality, which groups multiple devices together based on their proximity, allowing for a clear overview of your distributed sensor network.

To create a new device, you will need to provide the following attributes:

- Device Name
- Device Description
- Device Location Name
- Device Location Description
- Location Latitude
- Location Longitude

3.3 Sensor Types

You can access Sensor Types (referred to as "Sensors" in SensorThings terminology) by visiting the /sensors page. On this page, you'll find a table listing the available sensors, and you have the option to update or delete them as needed. To create a new sensor, you will be required to provide the following information:

- Sensor Name

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- Sensor Description
- Sensor Metadata

3.4 Measurement Property

Measurement Properties, which are alternatively referred to as "Observed Properties" in SensorThings terminology, are available for management through the `/observation_properties` page. This page presents a table containing essential information, including ID, name, description, and definition, for each Measurement Property. You have the flexibility to make updates or deletions to these Measurement Properties as needed. For the creation of a new Measurement Property, you will be required to provide the following mandatory details:

- Measurement Property Name
- Measurement Property Description
- Measurement Property Definition

3.5 Datastreams

Datastreams play a crucial role in the application as they establish connections between devices, sensor types, and measurement properties. You can access Datastreams through the `/datastreams` page. To create a new Datastream, the following information is mandatory:

- Datastream Name
- Datastream Description
- Datastream Observation Type
- Unit Of Measurement Name
- Unit Of Measurement Symbol
- Unit Of Measurement Definition
- Selection of a previous Sensor Type
- Selection of a previous Device
- Selection of a previous Measurement Property

3.6 Location

You can access Locations by navigating to the `/locations` page. Here, you will find a list of all the locations associated with devices. The information displayed includes the ID, Name, Description, Longitude, and Latitude of each location. Additionally, you have the option to visualize these location coordinates on a map for your convenience.

Regarding location management, historical locations are non-editable to maintain data integrity. For active locations, you have the flexibility to edit the details as needed. Furthermore, in the location detail page, you can create a new location for a device; this action automatically moves the current location into the historical record and establishes the new coordinates as the active location.

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3.7 Observations

You can access Observations by visiting the /observations page. Each observation is accompanied by its unique ID, the phenomenon time (indicating when it occurred), the result, and the associated datastream that it is linked to.

Section 4 Dataspace Specs

To access Dataspace Specs, you can simply click on "Quick Entry" within the "Data Space" tab located in the sidebar menu. The Dataspace Specs page provides two prominent buttons: "Stepper" and "Observations". The "Stepper" button leads to a page that allows you to efficiently create a sequence comprising Device, Sensor Type, Measurement Property, and Datastream in a single form. The "Observations" button redirects you to the Observations page.

Furthermore, this page displays valuable technical information, including the URL of the FROST Server instance in use, the FROST port, and general server details. To assist with authentication and resource monitoring, users can also view their Keycloak Client ID and Client Secret, as well as the total database memory currently being utilized by their dataspace. Additionally, it displays the counts of devices and datastreams currently available

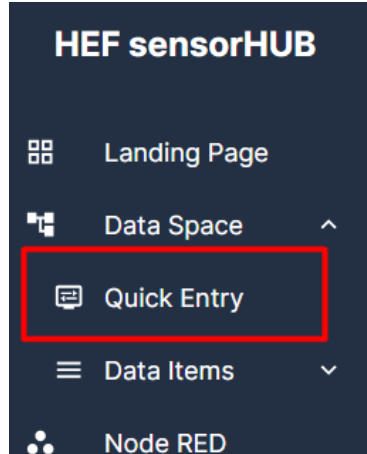


Figure 3 - Accessing Datspace Specs

4.1 Stepper

The "Stepper" feature is an incredibly efficient tool that enables the creation of a sequence of components all on one convenient page.

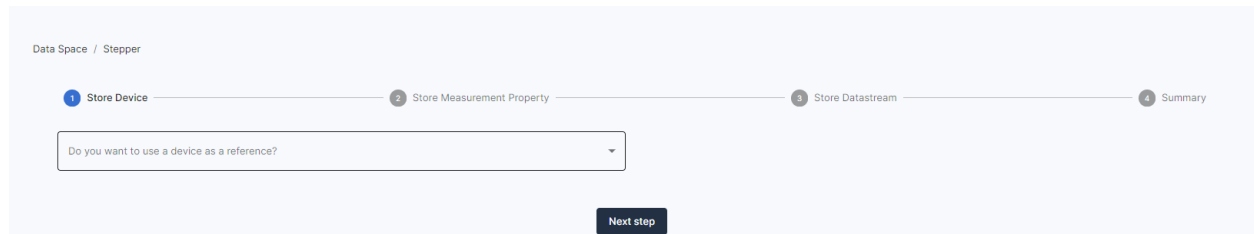
When it comes to the device section, users have the flexibility to either utilize an existing device along with its associated information or create a new device. It's worth noting that device names must be unique within the Stepper process.

For the Measurement Property section, users have the option to either select a previously defined measurement property or create a new one based on their needs.

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Moving on to the Datastream section, users can effortlessly generate a datastream by entering its name manually or simply by clicking "Generate Name."

This automated naming convention combines the device and measurement property, streamlining the process. Finally, users are presented with a summary page that allows them to double-check for any potential errors or oversights before proceeding. This comprehensive approach ensures accuracy and ease of use throughout the Stepper workflow.



The screenshot shows a web interface for the 'Stepper' process. At the top, it says 'Data Space / Stepper'. Below this is a horizontal progress bar with four numbered steps: 1. Store Device, 2. Store Measurement Property, 3. Store Datastream, and 4. Summary. Step 1 is highlighted with a blue circle. Below the progress bar is a dropdown menu with the text 'Do you want to use a device as a reference?'. To the right of the dropdown is a dark blue button labeled 'Next step'.

Figure 4 - Stepper Page

Section 5 Log Book

The Log Book serves as a centralized record for tracking project-related activities, such as sensor maintenance, movements, or other significant events. Users can access this feature by navigating to the /log_book page.

Viewing activities is facilitated by a table displaying recorded entries, including the ID, a description of the activity, the timestamp when the event occurred, and the timestamp when it was recorded in the system. To document a new activity, click the "Create" button; a dialog box will appear, allowing you to select the specific date and time of the event via the "Event Timestamp" field and provide a detailed description of the activity. Users can also maintain external backups or reports by using the "Export Page" or "Export All" buttons to download the log data.

Section 6 Audit Trail

The Audit Trail provides a transparent, automated history of all significant actions performed within your dataspace. Users can access this feature by navigating to the /audit_trail page.

The system automatically captures and records every critical activity, including the creation, modification, updating, and deletion of components such as devices, datastreams, and locations. Each entry in the Audit Trail provides essential details to ensure full traceability, including the timestamp of the event, the User ID of the individual who performed the action, the specific **Entity ID** associated with the component, the activity description, and the resulting status (e.g., "Success (201)").

For security and data integrity, the Audit Trail is a read-only record and cannot be manually modified by users. Additionally, to assist with reporting, compliance, or record-keeping, users can utilize the "Export CSV" button to download the audit logs for external analysis.

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Section 7 Contact Form

You can access the contact form by navigating to the /contact page. This form serves as a direct channel for users to reach out to the administrator at any time, whether they have questions or need assistance with an incident. Messages submitted through this form will be promptly forwarded to the administrator for a quick and efficient response.

Section 6 Knowledge Section

The Knowledge Section holds significant importance within our application, as it provides users with essential information regarding three critical components:

a) Node Red:

This section offers comprehensive insights into Node-RED, including valuable links to crucial resources and tutorials that can help users better understand and utilize Node-RED effectively.

b) Sensor Database

Here, users can access information and links related to the FROST server, as well as documentation from the Open Geospatial Consortium. This resource empowers users to explore and leverage the Sensor Database with confidence.

c) Web App

In this segment, users have the opportunity to download the manual for the HEF SensorHUB. This manual serves as a valuable guide, aiding users in navigating and making the most of the application's features and functionalities.